

Conditions, Loops & Iteration

- If statement is the most simple decision-making statement.
- If a certain condition is true then a block of statement is executed otherwise not.
- Syntax:- if condition:
 # Statements to execute if
 # condition is true
- Example:- i = 5
 if i < 9:
 print("5 is less than 9")

- What we do if condition False then we can add else statement.
- Syntax: if (condition):
 - # Executes this block if
 - # condition is true
 - else:
 - # Executes this block if
 - # condition is false
- Example: i = 5
 - if i < 9:
 - print(i,"is less than 9")
 - Else:
 - print(i,"is greater than 9")

- If-else takes only one condition, but what we do if we take multiple condition so we use elif, how let's see.
- Syntax: if (condition):
 statement
elif (condition):
 statement
else:
 Statement
- We can use multiple elif between if and else. Let's see example of it.

- Example:- i = 20
 if i == 10:
 print("i is 10")
 elif i == 15:
 print("i is 15")
 elif i == 20:
 print("i is 20")
 else:
 print("i is not present")

- We also use nested if-elif-else condition in Python, Nested means in block of if condition that must be another if condition.
- A nested if is an if statement that is the target of another if statement, similar to elif and else statement.
- Syntax:-if (condition1):
 - # Executes when condition1 is true
 - if (condition2):
 - # Executes when condition2 is true
 - else:
 - # Executes when condition2 is false
- else:
 - # Executes when condition1 is false

- if x < 0:
 x = 0
 """print('Negative changed to zero')"""
 if x == 0:
 print('Zero')
- elif x == 1:
 print('Single')
- if x == 0:
 print('Zero')
- else:
 print('More')

Basic Operators in Python

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- There are two logical operator which name is and operator and or operator
- a = True
b = False

```
if a or b:  
    print("hello1")
```

```
a = True  
b = True
```

```
if a and b:  
    print("hello2")
```

- In all programming languages, we use comparison operators like == (equal to), != (not equal to), < (less than), > (greater than), <= (less than equal to) and >= (greater than equal to)
- a = 10
b = 15
- print(a == b)
- print(a != b)
- print(a > b)
- print(a < b)
- print(a >= b)
- print(a <= b)

- There are some Arithmetic operators which perform on variables .
- Arithmetic operators are used to perform mathematical operations like addition, subtraction, multiplication and division.
- ```
a = 3
b = 2
print(a + b)
print(a - b)
print(a * b)
print(a / b)
print(a % b)
print(a ** b)
print(a // b)
```